

WALNEY ISLAND, United Kingdom

WMO: 032140

Lat: 54.1250N

Lon: 3.2581W

Elev: 47

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.0	31.0	22.4	16.9	33.0	24.4	18.5	33.9	36.5	42.6	33.4	43.1	5.9	10	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	73.4	64.8	69.7	62.6	67.0	61.2	65.7	71.5	63.7	68.1	62.2	65.8	10.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	88.1	67.9	61.9	82.9	65.7	60.6	79.1	64.3	30.5	71.5	29.1	68.3	28.0	65.9	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.8	27.3	24.6	DB	25.8	79.2	2.2	3.8	24.2	82.0	22.9	84.2	21.6	86.3	20.0	89.1	(4)
(5)			WB	24.6	70.1	2.3	3.7	23.0	72.7	21.7	74.9	20.4	76.9	18.8	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	41.2	41.2	43.5	47.6	52.7	57.0	60.6	60.7	57.3	52.5	46.6	42.5	(6)	
	DBStd	8.29	4.41	4.36	4.16	4.40	4.60	3.78	3.65	3.20	3.41	4.29	4.80	5.25	(7)	
	HDD50	1216	272	246	204	95	20	1	0	0	1	26	118	233	(8)	
	HDD65	5372	737	666	667	521	383	243	147	141	231	389	551	696	(9)	
	CDD50	1342	0	0	3	25	103	210	329	331	221	102	17	1	(10)	
	CDD65	23	0	0	0	0	1	3	11	7	1	0	0	0	(11)	
	CDH74	97	0	0	0	0	3	13	53	27	1	0	0	0	(12)	
	CDH80	9	0	0	0	0	0	0	6	3	0	0	0	0	(13)	
(14) Wind	WSAvg	12.2	13.6	12.7	12.4	11.2	11.3	11.1	11.0	11.3	11.7	13.2	13.2	13.4	(14)	
(15) Precipitation	PrecAvg	42.5	4.1	3.0	3.0	2.4	2.6	2.6	3.2	3.6	3.9	4.7	4.6	4.6	(15)	
	PrecMax	55.3	6.8	6.9	6.1	4.2	5.2	8.0	6.6	9.3	7.9	9.6	8.7	10.4	(16)	
	PrecMin	33.1	0.4	0.9	1.2	0.6	1.1	0.8	1.6	0.2	0.5	1.3	2.7	1.4	(17)	
	PrecStd	5.9	1.6	1.3	1.1	0.9	1.1	1.6	1.2	1.7	1.7	2.0	1.7	2.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.0	51.6	58.7	66.4	73.1	75.8	79.9	78.5	71.3	63.5	57.2	53.4	(19)	
		MCWB	49.2	47.9	50.3	55.0	62.7	65.5	66.9	69.0	64.1	58.6	55.4	51.7	(20)	
	2%	DB	49.3	49.4	53.4	60.3	67.9	69.5	73.6	71.3	67.1	60.9	55.7	52.0	(21)	
		MCWB	47.6	47.1	48.5	52.1	59.2	62.0	65.2	64.4	61.4	57.4	53.7	50.6	(22)	
	5%	DB	48.2	48.1	50.6	56.5	63.2	65.8	69.5	67.8	64.3	59.5	54.3	50.7	(23)	
		MCWB	46.5	46.2	47.1	50.6	56.2	59.3	63.3	62.2	59.7	56.6	52.2	49.0	(24)	
	10%	DB	47.2	47.0	48.8	53.8	59.6	63.3	66.5	65.5	62.5	58.3	53.0	49.3	(25)	
		MCWB	45.5	45.2	46.1	49.4	54.2	58.2	61.5	60.9	58.4	55.5	50.9	47.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	49.3	52.2	56.4	64.0	67.2	70.4	68.8	65.2	60.2	56.2	52.4	(27)	
		MCDB	50.6	50.3	56.0	63.9	71.8	73.8	75.9	74.1	69.6	62.2	56.7	52.9	(28)	
	2%	WB	48.1	47.9	49.4	53.3	59.8	62.5	66.0	65.1	62.5	58.5	54.3	51.0	(29)	
		MCDB	48.9	48.9	52.1	58.2	66.8	68.6	72.6	69.7	65.7	60.1	55.2	51.8	(30)	
	5%	WB	47.0	46.8	47.9	51.5	57.1	60.3	63.9	63.3	60.8	57.3	52.7	49.4	(31)	
		MCDB	47.9	47.8	49.9	55.5	61.7	64.8	68.6	66.7	63.1	59.0	53.9	50.5	(32)	
	10%	WB	45.9	45.6	46.8	50.0	55.1	58.8	62.1	62.0	59.3	56.0	51.4	48.0	(33)	
		MCDB	47.0	46.8	48.6	53.3	58.8	62.5	65.8	64.8	61.8	58.0	52.9	49.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.3	8.7	10.3	10.7	9.7	9.3	9.1	9.3	8.1	7.0	6.9	(35)	
		MCDBR	5.7	6.8	10.6	13.9	16.2	14.6	15.0	13.1	11.6	8.2	6.5	6.6	(36)	
	5% WB	MCWBR	6.1	6.0	7.5	9.0	10.6	9.4	9.4	8.4	8.0	6.7	6.8	7.0	(37)	
		MCWBR	5.8	6.2	9.0	12.1	14.3	12.4	13.0	11.0	10.0	7.4	6.3	6.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.330	0.341	0.368	0.387	0.391	0.388	0.390	0.376	0.365	0.350	0.326	0.317	(40)	
		taud	2.294	2.369	2.324	2.299	2.338	2.375	2.374	2.425	2.457	2.445	2.368	2.270	(41)	
	Ebn at Noon	Ebn at Noon	193	237	255	266	271	273	270	267	255	231	196	171	(42)	
		Edh at Noon	19	24	31	36	37	36	36	32	27	22	17	16	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	184	399	759	1197	1544	1580	1516	1230	871	471	230	141	(44)	
	RadStd	RadStd	22	44	66	113	117	151	154	86	71	50	24	14	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page